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General Certificate of Education (Adv. Level) Examination (Model), 2024

ဇီဝ විද්‍යාව I
Biology I

09 E I

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Two hours

- ● **Instructions:**
- *Answer all questions.
- *Write your Index Number in the space provided in the answer sheet.
- *Instructions are given on the back of the answer sheet. Follow them carefully.
- *In each of the questions from 1 to 50, pick one of the alternatives from (1), (2), (3), (4), (5) which is correct or most appropriate and mark your response on the answer sheet with a cross (×) on the number of the correct option in accordance with the instructions given on the back of the answer sheet.

1) Which is not a characteristic of organisms?

- 1) Adaptation
- 2) Reproduction
- 3) Movement
- 4) Heredity and Evolution
- 5) Metabolism

2) Which of the following statements is true about nucleotides?

- 1) Contain hexose sugar
- 2) Act as enzymes
- 3) Act as food reserves
- 4) Act as organic cofactors
- 5) Act as oxygen carriers

3) The ‘part-function’ relationship of a light microscope is correctly indicated in ,

- 1) Coarse adjustment - Focusing the specimen only under low power
- 2) Objective lens - Magnifies the specimen
- 3) High-power objective - Observation of a large number of cells
- 4) Fine adjustment - Focusing the specimen only under high power
- 5) Stage - To keep the microscope stable

4) Which of the following statements about water is correct?

- 1) Its hydrogen atoms are positively charged
- 2) Provides a biological medium for many organisms
- 3) Hydrogen bonds are very fragile in the liquid state
- 4) It is a small nonpolar, angular molecule
- 5) Small insects can walk on the surface due to the presence of high adhesive forces

5) Which statement is false about the electron transport chain?

- 1) NADH & FADH₂ produced in the early stages of aerobic respiration is oxidized by transferring electrons, through the electron transport chain
- 2) 28 ATP is formed in the electron transport chain of liver cells
- 3) Composed of a series of protein and non-protein molecules involving in the movement of electrons and protons across the cristae
- 4) Electron transport chain cannot be seen in bacteria
- 5) One molecule of FADH₂ produces 1.5 molecules of ATP by substrate-level phosphorylation

6) Which statement is false about the mitotic phase?

- 1) The nuclear envelope fragments during pro metaphase
- 2) The movement of the centrosomes towards opposite poles begins in prophase
- 3) Chromosomal arms of sister chromatids are attached by proteins called cohesin
- 4) Sister chromatids are separated at the centromere during anaphase
- 5) Chromosomes are attached to kinetochore microtubules during pro metaphase, metaphase & anaphase

7) Enzymes,

- 1) Catalyze any chemical reactions.
- 2) Its active sites are always complementary to the substrate
- 3) Are rigid structures
- 4) The shape of the active site changes when the temperature exceeds the optimum temperature
- 5) Bonds that form enzyme-substrate complex do not change when pH value changes.

8) Which of the following statements is false?

- 1) The first photosynthetic organisms appeared before 3.8 billion years ago.
- 2) The human lineage initiated 6-7 million years ago.

- 3) The earliest tetrapods formed about 365 million years ago
- 4) Fossils of the oldest known protists were dated as 1.2 billion years ago.
- 5) Recent studies related to alkaline vents show the abiotic synthesis of organic molecules.

9) Consider the following characteristics of fungi:

*In asexual reproduction, conidia are produced endogenously at the tip of the conidiophores which are specialized hyphae.

- * During sexual reproduction, ascospores are produced within asci.
- * During sexual reproduction, a sturdy structure called zygosporangium is produced.
- * Produce fruiting bodies called basidiocarps during sexual reproduction.

The characteristic features of *Penicillium* are:

- 1) A only
- 2) B only
- 3) A and B only
- 4) A, B, D only
- 5) All of the above

10) Incorrect statement about kingdom Plantae is:

- 1) All club mosses are homosporous and all spike mosses are heterosporous.
- 2) Root tissues of early vascular plants preserved in fossils resemble the stem tissues of living plants.
- 3) Seed plants are unique in retaining the megasporangium within the parent sporophyte.
- 4) Plants of phylum Cycadophyta produce flagellated sperms similar to those of seedless vascular plants.
- 5) Young sporophytes of phylum Bryophyta are not independent.

11) Which of the following statements is correct?

- 1) Organisms of phylum Chordata have a longitudinal, rigid rod called notochord located between digestive tube and nerve cord.
- 2) Although birds do not have color vision, they have an excellent eyesight.
- 3) Abdominal muscular heart can be seen in all chordates.
- 4) First appearance of little complex nervous and sensory system is first appeared in phylum Annelida.
- 5) Organisms of class Amphibia have a tympanic membrane found behind the eye.

12) The correct sequence of occurrence of following events is shown in:

- * A. First seed plant appeared.
- * B. Oldest fossils of eukaryotic cells appeared.
- * C. Flowering plants appeared and diversified.
- * D. Appearance of bipedal human ancestors.

Which one?

- 1) B, A, C, D
- 2) B, C, D, A
- 3) B, A, D, C
- 4) A, C, D, B
- 5) A, B, C, D

13) A characteristic of lateral meristems is:

- 1) Contributes to the replacement of the epidermis.
- 2) Contributes to the formation of young leaves.
- 3) Allow regrowth in damaged leaves.
- 4) Only located immediately inner to the epidermis.
- 5) Increase the length of plant stems and roots.

14) Which of the following statements is correct regarding the phenomena occurring during gravitropism?

- * A. Calcium ion concentration is higher in the cells of the lower side of the root.
- * B. Cells on lower side of the root grow rapidly.
- * C. Oxygen concentration in the cells of upper side of the root is low.
- * D. Statoliths are absent in the root cap.

Which one?

- 1) A, B 4) B, C
- 2) A, C. 5) A, B, C
- 3) A

15) What causes the lateral shoot growth in a plant due to trimming of its shoot tips?

- 1) Due to stimulation of gibberellin production.
- 2) Due to apical dominance by removal of lateral buds.
- 3) Removal of apical meristems reduces ethylene production and stimulates lateral branching.

4) Removal of apical meristems leads to increased production of auxin and stimulates the growth of lateral bud branches.

5) Growth of lateral branches due to reduced auxin due to removal of apical meristems.

16) Some characteristics of plant life cycles are given below:

A. Sporophytes are heterosporous and dioecious, and young leaves are fiddleheads.

B. The dominant plant contains xylem and phloem tissues, homosporous, gametophyte is photosynthetic.

C. Gametophyte is unisexual, Sporophyte depends on the gametophyte, Homosporous.

D. Gametophyte partially depends on the sporophyte, Produce biflagellated male gametes.

Which of the following shows A, B, C, D, E in sequence?

1) *Cycas*, *Selaginella*, *Pogonatum*, *Nephrolepis*, *Cycas*

2) *Selaginella*, *Cycas*, *Pogonatum*, *Nephrolepis*, *Cycas*

3) *Selaginella*, *Cycas*, *Nephrolepis*, *Pogonatum*, *Cycas*

4) *Cycas*, *Nephrolepis*, *Pogonatum*, *Selaginella*, *Cycas*

5) *Pogonatum*, *Selaginella*, *Nephrolepis*, *Cycas*

17) Which of the following statements is false?

1) The random motion of molecules from a place where it is more concentrated to a place where it is less concentrated, in the presence of external forces is called diffusion.

2) Diffusion can also take place across the membrane.

3) Metabolic energy is not used in diffusion.

4) Diffusion occurs due to a random motion.

5) O₂ and CO₂ can diffuse through the plasma membrane.

18) The false statement regarding plant growth substances is:

1) Natural & or synthetic organic compounds.

2) Active at very low concentrations.

3) Always act away from the place of production.

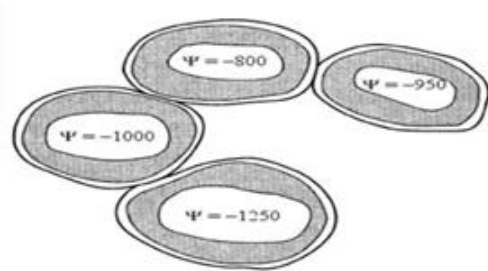
4) Transported through xylem and phloem.

5) Plant growth substances are important in responding to stimuli.

19) The false statement regarding guard cells is:

- 1) Possess specialized thickened walls.
- 2) Neighboring epidermal cells contribute to the turgor of guard cells.
- 3) In the presence of light - water potential reduces.
- 4) Always bean-shaped in angiosperms.
- 5) Cellulose microfibrills are abundant in the walls surrounding the stomatal pore.

20) The direction of movement of water in the cellular system is correctly indicated in the figure are correctly indicated in



- 1) D- $\rightarrow$ C, C- $\rightarrow$ B, B- $\rightarrow$ A.
- 2) C- $\rightarrow$ D, C- $\rightarrow$ B, B- $\rightarrow$ A.
- 3) C- $\rightarrow$ B, C- $\rightarrow$ D, A- $\rightarrow$ B.
- 4) A- $\rightarrow$ B, B- $\rightarrow$ C, C- $\rightarrow$ D
- 5) D- $\rightarrow$ C, B- $\rightarrow$ C, A- $\rightarrow$ B

21) Select the incorrect statement from the following statements.

- 1) Liver regulates the distribution of nutrients that reach the liver, to the rest of the body.
- 2) Excess glucose is stored as glycogen in the liver.
- 3) Vitamins D, E and copper are stored in the liver.
- 4) All the absorbed nutrients reach the liver.
- 5) Bile, which is stored in the gall bladder, is released into the duodenum.

22) Anemia occurs when which of the following vitamins/ minerals are deficient?

- 1) Vitamin B₁₂, Vitamin B₉, Iron, Calcium
- 2) Vitamin A, Vitamin E, Vitamin B₃, Vitamin B₇.
- 3) Magnesium, Pyridoxine, Cobalamin
- 4) Vitamin B₆, folic acid, Iron, Niacin
- 5) Vitamin A, Vitamin E, Vitamin B₁, Vitamin B₂

23) Select the incorrect statement regarding the pH value of body fluids.

- 1) pH of human blood is around 7.4.
- 2) Plasma ions and albumin plasma protein involve in regulating blood pH value.
- 3) Saliva is a neutral fluid.
- 4) Gastric juice is strongly acidic.

5) Bicarbonate ions present in gastric juice function as buffers.

24) The correct selection of epithelial tissue in different places of the body is:

- 1) Walls of alveoli - Pseudostratified ciliated epithelium.
- 2) Nasal cavity - Simple columnar ciliated epithelium.
- 3) Intestinal lining - Simple columnar non-ciliated epithelium.
- 4) Distal convoluted tubule of nephron - Stratified squamous epithelium.
- 5) Lining of oral cavity - Simple squamous epithelium.

25) Which response represents the correct sequence of passage of an atmospheric Oxygen molecule into the Alveoli?

- 1) Nose, nasal cavity, Larynx, pharynx, trachea, bronchi, bronchioles
- 2) Nose, nasal cavity, pharynx, larynx, bronchi, trachea, bronchioles
- 3) Nasal cavity, pharynx, trachea, bronchi, bronchioles, alveoli
- 4) Nose, Nasal cavity, pharynx, trachea, bronchi, bronchioles, respiratory bronchioles, alveoli
- 5) Nose, pharynx, trachea, bronchi, bronchioles, respiratory bronchioles, alveoli

26) Select the correct statement regarding the cardiac muscles?

- 1) Consists of long cylindrical cells with multinuclei
- 2) Possess intercalated discs to interconnect cells through their ends
- 3) Consists of structures corresponding to blood circulatory system within the walls
- 4) They consist of sarcomeres and are involuntary
- 5) Become fatigue only when contract rapidly

27) Which of the following represents the correct sequence of transmission of a nerve impulse which is transmitting along an axon of a neuron, through a chemical synapse?

- A. Diffusion of Ca^{2+} ions into the axon terminal
- B. Binding of neurotransmitters with the specific receptors
- C. Binding of synaptic vesicles with the presynaptic membrane
- D. Depolarization of the plasma membrane of the presynaptic cell
- E. Release of neurotransmitters into the synaptic cleft.

1) ACDEB

2 ADCBE

- 3) ABCDE
- 4) DACEB
- 5) DCAEB

28) Select the response with the common statement for both ovaries and testis

- 1) They produce the same type of hormones
- 2) Their development takes place within the abdominal cavity
- 3) The gametes produced by both are roughly the same in size
- 4) The gametes produced by both are the same in structure
- 5) The function performed by both the structures is regulated by the same type of hormones

29) Select the response which is most appropriate regarding (A) and (B) statements:

- A) Zygomatic bone helps for the movement of the mandible
- B) Zygomatic arch which is a facial bone, forms the zygomatic arch with the mandible

- 1) Both statements are correct and, B statement correctly describes by A
- 2) Both statements are correct and, B statement does not describe by A
- 3) A statement is correct while B is not correct
- 4) A statement is incorrect while B is correct
- 5) Both A and B statements are incorrect

30) How many Genotypes will obtain when an individual with the Genotype AaBb is subjected to a test cross?

- 1) 2. 4) 6
- 2) 3. 5) 8
- 3) 4

31) Which of the following conditions is not necessary to maintain a Hardy-Weinberg equilibrium in a population?

- 1) Absence of mutations
- 2) Absence of migration
- 3) Absence of natural selection
- 4) Absence of random mating
- 5) Size of the population is extremely large

32) Which response contains the correct sequence of the steps, occurring in translation process of the protein synthesis mechanism?

- A) Formation of the peptide bond between the methionine and the amino acid , brought by tRNA to the site A of the ribosome.
- B) 2 sub units of the ribosome combine to form the functional ribosome.
- C) Movement of the tRNA with the methionine to the E site
- D) Binding of the small subunit of the ribosome to mRNA and the initiator tRNA
- E) mRNA moves until the AUG start codon aligns with the P site of the ribosome.

- 1) BDEACF
- 2) BDCAFE
- 3) BDCEAF
- 4) DBCEAF
- 5) DBEACF



33) Select the incorrect statement regarding the agarose gel electrophoresis:

- 1) Agarose is a type of disaccharide obtained by a sea weed
- 2) Here the charged molecules are separated according to their mobility in an electric field
- 3) Here the stain, ethidium bromide can be used
- 4) Negatively charged DNA molecules migrate toward the anode through the gel
- 5) Here it not possible to separate a specific DNA sequence

34) Which of the following statement is with incorrect combination?

- 1) Endemic plants - *Dipterocarpus zeylanicus*, *Puntius nigrofasciatus*.
- 2) Indigenous specie - Lula, kitul
- 3) Relict species - Tuatara, *Lingula*
- 4) Endangered species - Elephant, Wesak orchid
- 5) Exotic species - Tilapia, Rubber

35) Which of the following convention/protocol is formulated in order to protect the ozone layer?

- 1) MARPOL convention
- 2) Basel convention
- 3) Ramsar convention
- 4) Montreal protocol

5) Kyoto protocol

36) Select the correct response based on the following statements:

* A) Cells of the gastrointestinal tract are destroyed by enterotoxines

* B) *Corynebacterium diptheriae* produces thermo-labile toxines

1) Both A and B are correct and A is affected by B

2) Both A and B are correct and A is independent of B

3) A is correct while B is not

4) A is incorrect while B is correct

5) Both A and B are incorrect

37) Several disease-causing agents are given below:

* A) Rubella virus

* B) *Neisseria meningitides*

* C) *Clostridium tetani*

* D) Hepatitis A virus

Which of the above agents affects to the nervous system?

1) A and B.

2) A and C.

3) A and D.

4) B and D.

5) B and C.

38) Select the correct statement regarding the viruses.

1) They are prokaryotic

2) They can be observed under the high power of light microscope

3) DNA and RNA present as genetic materials

4) They possess metabolic activities within living cells

5) They contain additional enzymes essential for protein synthesis process

39) Characteristic which cannot be seen in a culturable aquatic organism is,

1) Easy to breed

- 2) If it reaches sexual maturation relatively early, then it is advantageous
- 3) Ability to withstand the climate of the region in which it is cultured
- 4) Having tolerance to common diseases is advantageous
- 5) It should be an efficient converter of economical food stuff

40) Which of the following statements are incorrect regarding the tissue culture?

- 1) Most common benefit of tissue culture is that the cloning or, ability of mass production of genetically identical plants
- 2) Agar supplies certain nutrients to the culture medium
- 3) The Macro and micro nutrients in appropriate ratios make the inorganic salt component of the culture medium
- 4) Sucrose most commonly used organic energy source in tissue culture
- 5) MS medium is an example for the commercial media available for tissue culture

• For each of the questions 41 to 50, one or more of the responses is/are correct. Decide which response/responses is/are correct and then select the correct number.

- If only (A), (B) and (D) are correct (1)
- If only (A), (C) and (D) are correct (2)
- If only (A) and (B) are correct (3)
- If only (C) and (D) are correct (4)
- If any other response or combination of responses is correct (5)

41) Which of the following subcellular structure/structures are single membrane bounded?

- A) Mitochondrion
- B) Chloroplast
- C) Lysosomes
- D) Peroxisomes
- E) Centrioles

42) Which of the following/followings represent the correct 'organism - characteristic feature' combination?

- A) *Euglena* - Absence of cell wall
- B) *Amoeba* - Food vacuoles are present
- C) *Paramecium* - Lives in marine environments
- D) *Gelidium* - Flouredian starch is present as a storage food
- E) *Sargassum* - Lives in fresh water

43) Select the response/responses with the characteristic respiratory structures of Arthropods:

- A) Gills
- B) Lungs
- C) Book lungs
- D) Tracheal systems
- E) Body surface

44) Which of the following statement/statements are correct regarding guttation?

- A) Transpiration occurs at 100% of relative humidity causes the guttation
- B) In guttation water is removed from all herbaceous plants as droplets
- C) Guttation fluid differs from dew
- D) Guttation does not occur through stomata
- E) Guttation does not occur during the day time

45) Which of the following statement/statements are correct regarding the gaseous composition of the human body?

- A) Composition of oxygen in inhaled air is higher than that of carbondioxide
- B) Composition of Oxygen is higher than that of carbondioxide in pulmonary veins and systemic arteries
- C) Composition of oxygen in the alveolar capillaries is lower than that of the composition of carbondioxide.
- D) Composition of CO₂ is higher than that of O₂ within body tissues .
- E) Composition of O₂ in exhaled air is higher than that of CO₂

46) Which of the following combination/combinations are correct regarding innate immunity?

- A) Monocytes - Ingest foreign particles
- B) Interferons - Stimulate the secretion of antiviral proteins
- C) Natural killer cells - Provides specific defense mechanism
- D) Complimentary proteins - Lysis of invading cells by biochemical reactions

E) Neutrophils - Secrete the chemicals which interfere with the viral replication

47) Which of the following/followings are secreted into the filtrate of the nephron?

- A) H⁺ ions
- B) Creatinine
- C) Excess glucose
- D) Excess K⁺ ions
- E) Na⁺ ions

48) Which of the following statement/statements are correct regarding mutations?

- A) They occur spontaneously within living cells
- B) They are important for the evolution of organisms
- C) Always transmit to the next generation during reproduction
- D) They can occur during meiotic division
- E) It doesn't alter the chromosomal number in a cell

49) Which of the response/responses consist of the plants in the, biomes with decreasing order of their altitudes?

- A) *Manilkara hexandra*, *Dichrostachys cineria*, *Halodule*
- B) *Cymbopogon nardus*, *Salvinia*, *Spinifex littoreus*
- C) *Themeda tremula*, *Arundinella villosa*, *Aponogeton sp*
- D) *Cinnamomum ovalifolium*, *Mesua ferrea*, *Calophyllum walkeri*
- E) *Calotropis gigantean*, *Terminalia chebula*, *Chrysopogon nodulibarbis*

50) Which of the following disease/diseases commonly occur among the fresh water aquatic fish species by bacteria?

- A) Fish white spot disease
- B) Gill and skin infestation
- C) Haemorrhagic septicaemia
- D) Columnaris disease
- E) Gill rot and fin rot



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Pre Medicine Association Kurunegala 2023

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General Certificate of Education (Adv. Level) Examination (Model), 2024

ජීව විද්‍යාව II
Biology II

09 E II

පැය තුනයි
Three hours

01)

(A)

(i) Mention 2 living characteristics shown by non-living organisms.

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(ii) What is Adaptation?

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(iii) Name 2 nitrogen containing polymers found in living organisms. Give 2 functions of each polymer named.

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(iv) State the Carbon dioxide acceptor(s) at following photosynthesis mechanisms.

C3 -

C4 -

(v) Give 2 products of light reaction which are utilized in the Calvin cycle of photosynthesis.

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(vi) Write the relevant phase of the eukaryotic cell cycle in which the following incidents take place:

a) Condensation of chromosomes

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b) DNA replication -

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c) Arrangement of chromosomes at the center of the cell -

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(B)

(i) Define Protocell.

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(ii) Mention 4 characteristics of protocells.

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(iii) Complete the following dichotomous key.

(Earthworm, Clam, Brittle Star, Sea anemone)

Radially symmetrical body -

Not having a radially symmetrical body -

Exoskeleton present -

Exoskeleton absent -

Anus present -

Anus absent -

(C)

(i) Following are some features regarding plant reproduction.

Indicate the relevant letters of plant named (A-E) for each of the following characteristics.

(A) *Pogonatum* (B) *Nephrolepis* (C) *Selaginella* (D) *Cycas* (E) Angiosperms

- (i) Photosynthetic gametophyte -
- (ii) Dioecious gametophyte -
- (iii) Heterosporous -
- (iv) Endosperm -
- (v) Sori -

(ii)Mention 5 factors affecting the rate of transpiration.

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(iii)Briefly explain seed germination.

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02)

(A)

(i)Define blood pressure.

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(ii)Mention two factors responsible for variation of blood pressure of an individual.

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(iii)What is meant by hypertension?

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(iv) How many days is the life span of a red blood cell?

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(B)

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[illegible]

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(iv) Name a water-soluble vitamin and a mineral element which is stored in the liver.

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(v) Mention 3 functions of protein in the diet.

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(C)

(i) What is immunity?

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(ii) Name 4 internal defense mechanisms in innate immunity.

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(iii) Name 2 autoimmune diseases caused due to cell-mediated immune response.

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(iv) Give 3 examples for antigen-presenting cells.

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(v) What are allergens?

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(vi) Name 3 common allergens.

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03) A)

i) Describe the term 'Water potential' and Write it in the form of equation and define its terms.

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ii) Describe the reason to present a positive pressure potential inside a living cell

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iii) What is meant by sugar source in phloem translocation?

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iv) Write a deficiency symptoms for each of the followings, when a plant possesses deficiency in following minerals

a)Cl -

b)Cu -

c)K -

V) How does cross pollination help in survival and evolution of a species

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B)

i) a) What is meant by Epigenetic inheritance?

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b) What are the factors which result in the above phenomenon?

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ii) a) What is the basic unit of heredity?

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b) Write the main characteristic features of it

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iii) State the Dominant/recessive nature of the below given, common Mendelian characters in humans

a) Attached ear lobes -

b) Absence of Dimples on cheeks -

c) Rolling of tongue -

iv) When a long bean plant with purple flowers and long pods are crossed with another plant with red flowers and round pods, all the F₁ progeny are with purple flowers and long pods. Followings are the results of F₂ generation obtained when F₁ progeny are subjected to self pollination.

Long purple – 2350

Round purple – 260

Long red -240

Round red -2150

Purple (P),

Red (p)

Long (L)

Round (l)

a) State the genotype of the 2 parent plants

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b) Write the types of gametes produced by parent plants

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c) State the genotype of F₁ plants

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d) Calculate percentage of the heterozygotes in the F₂ generation

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e) Is that the above hybridization is in accordance with the Mendel's laws

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f) State a reason to have the F₂ generation with more of the parental phenotypes and with the less number of other phenotypes

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C)

a) What is the instrument used to measure the rate of transpiration in plants?

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b)

i) Draw a labeled diagram of this instrument

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li) What is the main assumption used in this experiment

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ii) Write the changes that can be made in the above experiment to determine the effect of the following factors in Transpiration

a) Humidity –

b) Wind-

04)A)

ii) Mention 3 characteristics of DNA that facilitates DNA for acting as vital genetic material in organisms

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iii) What is calling proofreading activity of DNA polymerase?

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iv)

a) What are DNA libraries?

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b) Mention 2 types of DNA libraries

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v) State 3 types of gene mutations and explain how each of them occurs

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vi)

a) Who is called a genetically modified organism?

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b) Mention 3 uses of GMOs used to increase nutritional value of crops

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(B)

i) Name 3 things that make up the niche of an organism

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ii) What is a food web?

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iii) Mention the 3 types of ecological pyramids

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iv) What is a biome?

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v) In Chaparral,

- a) Average annual rainfall -
- b) Average temperature -
- c) Extended areas -
- d) Adaptations shown by plants -

vi)

a) What is the principal goal of conservation?

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b) Mention the two ways how conservation can be done and write an example for each

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(C)

i) Name the 3 basic shapes of bacteria

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ii) Name 2 chemical sterilizing agents

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iii)

a) What is called a culture medium?

.....
b) Name two common culture media and their contents

.....
iv) What is called pathogenicity?

.....
v) Write an example each for Antiseptics and disinfectants

.....
vi) Explain bioremediation and write an example issue

.....
(D)

i) Define 'Aquarium'

.....
.....
ii) Mention 2 activities that should be done for nightly maintaining a home aquarium?

.....
.....
iii) Name two warning signs of dengue

.....
iv) Write two major properties of Human Embryonic Stem Cells (hESCs)

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Part B – Essay

●Instructions

*Answer four questions only.

*Each question carries 150 marks.

01.) a.) Describe general characteristics of enzymes and State relationship between enzyme and activation energy.

b.) Explain the mechanisms that regulate enzyme. activity in a cell.

02.) a.) Explain the transpiration process of water loss in plants.

b) Describe the factors affecting the rate of transpiration.

03) a) Describe the structure of human outer and middle ear.

b) Describe the role of the human ear in maintaining equilibrium.

04) a) Describe the structure of a typical vertebra.

b) Describe the accepted mechanism of striated muscle contraction.

05) a) Describe the use of antibiotics in controlling microbial diseases.

06) Describe the interactions of soil microorganisms relevant to plant growth.

07) Write short notes on the following.

a) Loose connective tissue

b) Genetic code.

c) Embryonic stem cells.

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